

IILM UNIVERSITY

SECURE MULTI-TENANT LEGAL TECHNOLOGY PLATFORM DEVELOPMENT

Student	AMRITESH GAUTAM
Enrollment No.	CS2341800
Program & Semester	B.Tech CSE - Semester VII
Organization	Clearpaper
Role	Independent Software Developer

Batch: 2023–27 | Section: 4CSE3 | Academic Year: 2026–27

1. Engagement & Organization

Organization

Early-stage legal technology startup based in London, United Kingdom, developing a SaaS platform for legal workflows.

Engagement

Independent Software Developer
Remote • Part-time
Started January 2026
Extended until October 2026

Development Context

Small development team working on an actively evolving product with emphasis on security, reliability, maintainability and rapid iteration.

Focus of the engagement

- Contributing to a secure legal-technology SaaS platform.
- Working across backend, frontend, data, document infrastructure and integrations.
- Supporting selected AI-assisted workflows while keeping unreleased product details confidential.

2. Problem Statement & Objectives

Problem Statement

Legal professionals handle large volumes of sensitive documents and matter-related information. Conventional file-sharing may not provide sufficient structure around matters, roles, permissions, versions and auditability.

Project Objectives

- Secure multi-tenant SaaS development
- Backend services, APIs and data management
- Authenticated, matter-based frontend workflows
- Secure document storage and version management
- Permission-aware application behaviour
- Integrations and workflow automation
- Selected AI-assisted activity and time-tracking workflows
- Reliability, usability and maintainability

3. Scope, Technologies & Tools

Project Scope

Full-stack development across backend APIs, database-backed logic, frontend interfaces, authentication and authorization, document management, integrations and selected automation/AI-assisted workflows.

Engineering Responsibilities

Cloud-backed storage, document-processing pipelines, permission-aware operations, data consistency, iterative development and adaptation to changing requirements.

TypeScript / Node.js
Backend

Express
REST APIs

Next.js / React
Frontend

PostgreSQL / Prisma
Data

AWS S3
Object storage

Microsoft Graph
Integrations

PDF / Text Libraries
Document processing

Git / GitHub
Source control

4. High-Level Architecture & Methodology

Web Application
User interface

Backend API
Auth, validation & business logic

Database
Structured application data

Object Storage
Secure document files

Document Processing
Text extraction

External Integrations
Selected external services

Iterative methodology

1
Requirement understanding

2
Design

3
Implementation

4
Integration

5
Verification

6
Iteration

5. Outcomes & Key Takeaways

- Practical understanding of production-oriented full-stack software development.
- Hands-on experience with secure multi-tenant workflows, authorization and document management.
- Development experience with TypeScript, Node.js, Express, Next.js, React, PostgreSQL and Prisma.
- Exposure to cloud object storage, document processing and external API integrations.
- Experience contributing to AI-assisted workflows in an evolving product environment.
- Improved ability to work independently, communicate technical decisions and deliver features within a small team.

Thank You